ANSWERS OF MODEL TEST PAPER 2

INTERMEDIATE: GROUP – II

PAPER – 6: FINANCIAL MANAGEMENT & STRATEGIC MANAGEMENT

PAPER 6A : FINANCIAL MANAGEMENT

PART I – Case Scenario based MCQs

1. **1.** **(d)** **14.99%**

B = retention ratio=0.6, r=return on equity=20%, DPS=D0=20 x 0.4= 8,

MPS = P0 = EPS x PE = 20 x 15=300

G = b.r =0.6 x 20% = 12%

D1 = D0(1+g) = 8 (1.12) = 8.96

Ke = D1/P0 + g = 8.96/300 + 0.12 = 14.99%

2. **(c)** **90.58**

Price of debentures= PV of future cash flows for investor discounted at their yield

= 8 x PVAF(9.5%,10 years)+ 100 x PVF(9.5%, 10 years)

= 8 x 6.2788 + 100 x 0.4035

=50.2304 + 40.35

=90.58

3. **(a)** **7.64%**

NP = 90.58 x 96% = 86.96, RV = 100, Interest = 8, t = 0.27, n = 10

$$Kd = \frac{Int(1-t)+(RV-NP)/n}{(RV+NP)/2}$$

$$= \frac{8(1-0.27)+(100-86.96)/10}{(100+86.96)/2}$$

= 7.64%

4. **(b)** **9.77%**

$$Kp = \frac{PD+(RV-NP)/n}{(RV+NP)/2}$$

$$= \frac{100+(1100-1050)/10}{(1100+1050)/2}$$

= 9.77%

5. **(a)** **10.52%**

	Existing	Total	Additional	
Equity Funds	1,60,00,000	2,00,00,000	40,00,000	

Preference Shares		24,00,000	24,00,000	
Debt		56,00,000	56,00,000	
	1,60,00,000	2,80,00,000	1,20,00,000	
Capital gearing =	0.4			
(PSC + Debt)/Equity =	0.4			
(Total Funds -Equity)/ Equity = 0.4				
(2.8 crores-Equity)/ equity = 0.4				
Equity =	2 crores			
Weighted avg cost of marginal capital		**Weights**	**Cost**	**W.C**
Equity Funds	40,00,000	0.333333333	14.99%	5.00%
Preference Shares	24,00,000	0.2	9.77%	1.952%
Debt	56,00,000	0.466666667	7.64%	3.565%
Total	**1,20,00,000**			**10.52%**

2. **(a)** **9.74%**

$$K_d = \frac{I(1-t) + \frac{RV - NP}{n}}{\frac{RV + NP}{2}}$$

$$K_d = \left[\frac{10(0.7) + \frac{110 - 85}{10}}{97.5} \right]$$

= 9.50/97.5 = 9.74%

3. **(c)** **2.5**

Margin of safety = (sales – BEP sales)/sales x 100

= 40%

Degree of operating leverage = 1/MOS

= 1/40% = 2.5

4. **(a)** **20%**

$$\text{Payback Reciprocal} = \frac{\text{Average annual cash in flow}}{\text{Initial investment}}$$

$$= \frac{₹\ 4{,}000 \times 100}{₹\ 20{,}000} = 20\%$$

PART II – Descriptive Questions

1. (a) (i) Working Notes:

(i) Computation of Annual Cash Cost of Production	**(₹)**
Material consumed	12,00,000
Wages	9,60,000
Manufacturing expenses	12,00,000
Total cash cost of production	**33,60,000**
(ii) Computation of Annual Cash Cost of Sales:	**(₹)**
Total Cash cost of production as in (i) above	33,60,000
Administrative Expenses	3,60,000
Sales promotion expenses	1,20,000
Total cash cost of sales	**38,40,000**
Add: Gross Profit @ 20% on sales (25% on cost of sales)	9,60,000
Sales Value	**48,00,000**

Statement of Working Capital requirements (cash cost basis)

	(₹)	**(₹)**
A. Current Assets		
Inventory:		
- Raw materials $\left(\frac{₹\ 12,00,000}{12\text{ months}} \times 2\text{ months}\right)$	2,00,000	
- Finished Goods $\left(\frac{₹\ 33,60,000}{12\text{ months}} \times 2\text{ months}\right)$	5,60,000	
Receivables (Debtors) $\left(\frac{₹\ 38,40,000}{12\text{ months}} \times 3\text{ months}\right)$	9,60,000	
Sales Promotion expenses paid in advance $\left(\frac{₹1,20,000}{12\text{ months}} \times 1\text{ month}\right)$	10,000	
Cash balance	1,00,000	18,30,000
Gross Working Capital		18,30,000

B. Current Liabilities:		
Payables:		
- Creditors for materials $\left(\frac{₹\,12,00,000}{12\,months} \times 2\ months\right)$	2,00,000	
Wages outstanding $\left(\frac{₹\,9,60,000}{12\,months} \times 1\ month\right)$	80,000	
Manufacturing expenses outstanding $\left(\frac{₹\,12,00,000}{12\,months} \times 1\ month\right)$	1,00,000	
Administrative expenses outstanding $\left(\frac{₹\,3,60,000}{12\,months} \times 1\ month\right)$	30,000	4,10,000
Net working capital (A - B)		14,20,000
Add: Safety margin @ 15%		2,13,000
Total Working Capital requirements		16,33,000

(b) (i) Calculation of market price per share

According to Miller – Modigliani (MM) Approach:

$$P_o = \frac{P_1 + D_1}{1 + K_e}$$

Where,

Existing market price (P_o) = ₹ 600

Expected dividend per share (D_1) = ₹ 40

Capitalization rate (k_e) = 0.20

Market price at year end (P_1) = ?

a. If expected dividends are declared, then

600=(P1+40)/(1+0.2)

600 x 1.2 = P1+40

P1 = 680

b. If expected dividends are not declared, then

600 = (P1+0)/(1 + 0.2)

600 x 1.2 = P1

P1= 720

(ii) Calculation of number of shares to be issued

	(a)	(b)
	Dividends are declared (₹ lakh)	**Dividends are not Declared (₹ lakh)**
Net income	1500	1500
Total dividends	(400)	-
Retained earnings	1100	1500
Investment budget	2000	2000
Amount to be raised by new issues	900	500
Relevant market price (₹ per share)	680	720
No. of new shares to be issued (in lakh) (₹ 900 ÷ 680; ₹ 500 ÷ 720)	1.3235	0.6944

(iii) Calculation of market value of the shares

	(a)	(b)
Particulars	**Dividends are declared**	**Dividends are not Declared**
Existing shares (in lakhs)	10.00	10.00
New shares (in lakhs)	1.3235	0.6944
Total shares (in lakhs)	11.3235	10.6944
Market price per share (₹)	680	720
Total market value of shares at the end of the year (₹ in lakh)	11.3235 × 680 = 7,700 (approx.)	10.6944 × 720 = 7,700 (approx.)

Hence, it is proved that the total market value of shares remains unchanged irrespective of whether dividends are declared, or not declared.

(c) Calculation of Cash Flow after Tax

	Year 1	Year 2	Year 3	Year 4	Year 5
Capacity	50%	65%	80%	100%	100%
Units	1,50,000	1,95,000	2,40,000	3,00,000	3,00,000
Contribution p.u.	360	360	360	360	360
(600 x 60%)					
Total Contribution	5,40,00,000	7,02,00,000	8,64,00,000	10,80,00,000	10,80,00,000
Less: Fixed Asset	2,00,00,000	3,50,00,000	5,00,00,000	5,00,00,000	5,00,00,000
Less: Depreciation (W.N.)	4,00,00,000	2,40,00,000	1,44,00,000	86,40,000	51,84,000

PBT	(60,00,000)	1,12,00,000	2,20,00,000	4,93,60,000	5,28,16,000
Less: Tax	(18,00,000)	33,60,000	66,00,000	1,48,08,000	1,58,44,800
PAT	(42,00,000)	78,40,000	1,54,00,000	3,45,52,000	3,69,71,200
Add: Depreciation	4,00,00,000	2,40,00,000	1,44,00,000	86,40,000	51,84,000
CFAT	3,58,00,000	3,18,40,000	2,98,00,000	4,31,92,000	4,21,55,200

Calculation of NPV

Year	Description	Cash Flow	PVF @12%	PV
0	Initial Investment	(10,00,00,000)	1	(10,00,00,000)
0	WC introduced	(1,50,00,000)	1	(1,50,00,000)
3	WC introduced	(2,00,00,000)	0.7118	(1,42,36,000)
1	CFAT	3,58,00,000	0.8929	3,19,65,820
2	CFAT	3,18,40,000	0.7972	2,53,82,848
3	CFAT	2,98,00,000	0.7118	2,12,11,640
4	CFAT	4,31,92,000	0.6355	2,74,48,516
5	CFAT	4,21,55,200	0.5674	2,39,18,860
5	WC released	3,50,00,000	0.5674	1,98,59,000
5	Scrap Sale	2,00,00,000	0.5674	1,13,48,000
	Net Present Value			**3,18,98,684**

Working Notes (W.N.)

Calculation of Depreciation

Year	Opening WDV	Depreciation	Closing WDV
1	10,00,00,000	4,00,00,000	6,00,00,000
2	6,00,00,000	2,40,00,000	3,60,00,000
3	3,60,00,000	1,44,00,000	2,16,00,000
4	2,16,00,000	86,40,000	1,29,60,000
5	1,29,60,000	51,84,000	77,76,000

2. (a) Income statement

Particulars		P	Q
		(₹)	(₹)
	Sales	50,00,000	48,00,000
(–)	Variable Cost	33,50,000	24,00,000
	Contribution	16,50,000	24,00,000
	Fixed Cost	8,25,000	16,00,000
	EBIT	8,25,000	8,00,000
(–)	Interest	5,50,000	6,00,000
	EBT	2,75,000	2,00,000

(–)	Tax	82,500	60,000
	EAT	1,92,500	1,40,000
(÷)	No. of Shares	1,00,000	70,000
	EPS	**₹ 1.93**	**₹ 2.00**

Working Note :

1. Financial Leverage	**=**	$\frac{\text{EBIT}}{\text{EBT}}$	**=**	$\frac{\text{EBIT}}{\text{(EBIT – Int.)}}$

Let the EBIT be X

	P	Q
	3 = X/(X – 5,50,000) 3(X – 5,50,000) = X 3X – 16,50,000 = X 2X = 16,50,000	4 = X/(X – 6,00,000) 4(X – 6,00,000) = X 4X – 24,00,000 = X 3X = 24,00,000
	X = 8,25,000	**X = 8,00,000**

2. Operating Leverage **=** Contribution/EBIT

Let the Contribution be X

	P	Q
	2 = X/8,25,000	3= X/8,00,000
	X = 16,50,000	**X = 24,00,000**

3. Sales

Let the Sales be 100

Sales – Variable Cost = Contribution

	P		**Q**
Contribution **=**	100 – 67	**=**	100 – 50
=	33	**=**	50
Sales **=**			
	P		**Q**
For 33	= 16,50,000	For 50	= 24,00,000
For 100	**= 50,00,000**	For 100	**= 48,00,000**

(b) Expected return on capital employed

Capital Employed = Debt + Equity

= (₹ 63,00,000 + ₹ 54,00,000) + (₹ 70,00,000 + ₹ 1,30,00,000)

= ₹ 3,17,00,000

Return on capital employed/ROI = $\left(\frac{\text{EBIT}}{\text{Capital employed}}\right)$ x 100

At present:

$$= \left(\frac{54,00,000}{3,17,00,000}\right) \text{ x } 100$$

= 17.03%

Now company expects 2% more as ROI

So, Expected ROI = 17.03% + 2%

= 19.03%

Proposed EBIT = Proposed Capital Employed x Return on capital employed

= (₹ 3,17,00,000 + ₹ 50,00,000) x 19.03% = ₹ 69,84,010

(i) Market Price per Share:

Particular	**Financial Options**	
	Option – I **12% term loan of ₹ 50,00,000**	**Option II** **1,00,000 equity shares @ ₹ 20 and 11% debentures of ₹ 30,00,000**
	(₹)	**(₹)**
EBIT	69,84,010	69,84,010
Less: Interest		
- 10% on old debentures	6,30,000	6,30,000
- 11% on new debentures	-	3,30,000
- 12% on old term loan	6,48,000	6,48,000
- 12% on new term loan	6,00,000	
Total Interest	18,78,000	16,08,000
EBT	51,06,010	53,76,010
Less Tax @ 30%	15,31,803	16,12,803
EAT	35,74,207	37,63,207
No. of equity shares	7,00,000	8,00,000
Earnings per share	**5.11**	**4.70**
P/E ratio	10	10
Market Price per Share = EPS x P/E ratio	**51.06**	**47.04**

(ii) Recommendation:

The option I is better and may be opted as both EPS and MPS are higher.

3. (a) Inventory Turnover = $\frac{\text{Inventory}}{\text{COGS}} \times 365 = \frac{38,60,000 \times 365}{76,40,000} = 184.41$ days = 185 days (apx)

Receivables Turnover = $\frac{\text{Receivables}}{\text{Sales}} \times 365 = \frac{39,97,000 \times 365}{1,25,00,000} = 116.71$ = 117 days (apx)

Equity to Reserves = 1

Reserves = 1 x 30,00,000 = 30,00,000

Projected profit = 30,00,000 - 18,00,000 = 12,00,000

Net Profit Margin = 15%

12,00,000/ Sales = 0.15

Sales = 80,00,000

Gross Profit = 80,00,000 x 50% = 40,00,000

COGS = 80,00,000 - 40,00,000 = 40,00,000

Projected Debtors Turnover = 100 days = $\frac{\text{Closing Receivables}}{\text{Sales}} \times 365$

$$100 = \frac{\text{Closing Receivables}}{80,00,000} \times 365$$

$$\text{Closing Receivables} = \frac{80,00,000 \times 100}{365} = 21,91,781$$

Projected Closing Inventory = 70% of opening inventory = 70% of 38,60,000 = 27,02,000

Projected Creditor Turnover= 100 days = $\frac{\text{Closing Creditors}}{\text{COGS}} \times 365$

Closing Creditors = $\frac{\text{COGS}}{365} \times 100$

Closing Creditor = $\frac{40,00,000}{365} \times 100 = 10,95,890$

Equity Share Capital + Reserves = 30,00,000 + 30,00,000 = 60,00,000

Long Term Debt to Equity = 0.5

$$\frac{\text{LTD}}{60,00,000} = 0.5$$

Long Term Debt = 0.5 x 60,00,000

Long Term Debt = 30,00,000

(b) Financial Instruments in the International Market

Some of the various financial instruments dealt with in the international market are:

(a) Euro Bonds

(b) Foreign Bonds

(c) Fully Hedged Bonds

(d) Medium Term Notes

(e) Floating Rate Notes

(f) External Commercial Borrowings

(g) Foreign Currency Futures

(h) Foreign Currency Option

(i) Euro Commercial Papers.

4. (a) Inter-relationship between Investment, Financing and Dividend Decisions: The finance functions are divided into three major decisions, viz., investment, financing and dividend decisions. It is correct to say that these decisions are inter-related because the underlying objective of these three decisions is the same, i.e. maximisation of shareholders' wealth. Since investment, financing and dividend decisions are all interrelated, one has to consider the joint impact of these decisions on the market price of the company's shares and these decisions should also be solved jointly. The decision to invest in a new project needs the finance for the investment. The financing decision, in turn, is influenced by and influences dividend decision because retained earnings used in internal financing deprive shareholders of their dividends. An efficient financial management can ensure optimal joint decisions. This is possible by evaluating each decision in relation to its effect on the shareholders' wealth.

The above three decisions are briefly examined below in the light of their inter-relationship and to see how they can help in maximising the shareholders' wealth i.e. market price of the company's shares.

***Investment decision**:* The investment of long term funds is made after a careful assessment of the various projects through capital budgeting and uncertainty analysis. However, only that investment proposal is to be accepted which is expected to yield at least so much return as is adequate to meet its cost of financing. This have an influence on the profitability of the company and ultimately on its wealth.

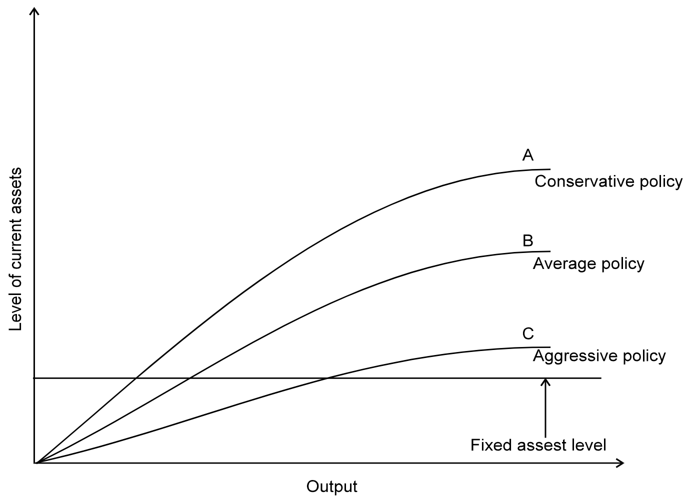
***Financing decision**:* Funds can be raised from various sources. Each source of funds involves different issues. The finance manager has to maintain a proper balance between long-term and short-term funds. With the total volume of long-term funds, he has to ensure a proper mix of loan funds and owner's funds. The optimum financing mix will increase return to equity shareholders and thus maximise their wealth.

Dividend decision*:* The finance manager is also concerned with the decision to pay or declare dividend. He assists the top management in deciding as to what portion of the profit should be paid to the shareholders by way of dividends and what portion should be retained in the business. An optimal dividend pay-out ratio maximises shareholders' wealth.

The above discussion makes it clear that investment, financing and dividend decisions are interrelated and are to be taken jointly keeping in view their joint effect on the shareholders' wealth.

(b) Liquidity versus Profitability Issue in Management of Working Capital

Working capital management entails the control and monitoring of all components of working capital i.e. cash, marketable securities, debtors, creditors etc. Finance manager has to pay particular attention to the levels of current assets and their financing. To decide the level of financing of current assets, the risk return trade off must be taken into account. The level of current assets can be measured by creating a relationship between current assets and fixed assets. A firm may follow a conservative, aggressive or moderate policy.



A conservative policy means lower return and risk while an aggressive policy produces higher return and risk. The two important aims of the working capital management are profitability and solvency. A liquid firm has less risk of insolvency i.e. it will hardly experience a cash shortage or a stock out situation. However, there is a cost associated with maintaining a sound liquidity position. So, to have a higher profitability the firm may have to sacrifice solvency and maintain a relatively low level of current assets.

(c) Concept of Discounted Payback Period

Payback period is time taken to recover the original investment from project cash flows. It is also termed as break even period. The focus of the analysis is on liquidity aspect and it suffers from the limitation of ignoring time value of money and profitability. Discounted payback period considers present value of cash flows, discounted at company's cost of capital to estimate breakeven period i.e. it is that period in which future discounted cash flows equal the initial outflow. The shorter the period, better it is. It also ignores post discounted payback period cash flows.

OR

(c) **Concept of Indian Depository Receipts:** The concept of the depository receipt mechanism which is used to raise funds in foreign currency has been applied in the Indian capital market through the issue of Indian Depository Receipts (IDRs). Foreign companies can issue IDRs to raise funds from Indian market on the same lines as an Indian company uses ADRs/GDRs to raise foreign capital. The IDRs are listed and traded in India in the same way as other Indian securities are traded.